

The Downside Deviation Quadratic Programming for Stock Portfolio Optimisation: An Empirical Study of Shariah and Conventional Indices in Indonesia

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Abstract

The Quadratic Programming (QP) for portfolio optimisation may yet be improved to generate better results on the risk. This study presents the Downside Deviation Quadratic Programming (DDQP) to optimise the risk of portfolio as a refinement of QP. The data deals with the price of stocks listed in Jakarta Islamic Index and IDX30 Indonesia for a definite interval. The selection of portfolio for all the stocks considered the sectoral approach. Upon selection, the DDQP model was constructed and applied to the selected portfolio before benchmarking to QP. The results showed that the portfolio group 1 had the best risk on the shariah platform, while the portfolio group 7 was superior to conventional. Additionally, the empirical analysis revealed that ten scenarios can be inferred based on the DDQP as it is consistently stable in producing a lower risk portfolio than the QP. Lastly, heuristic Pattern Search also verified the results of DDQP.

Keywords: Portfolio selection, portfolio optimisation, risk, quadratic programming, downside deviation quadratic programming, pattern search.

JEL classification: C44, C61, G32

1. Introduction

Optimisation models play a significant part in financial decision problems. Various complex financial problems such as capital allocation, risk-return management, stocks portfolio, and bond can be solved efficiently by way of modern computational optimisation methods (Elton et al., 2014). An optimised stock portfolio reflects to the group of stocks with the efficient result. Having said that, investors who financially literate often rational and prefer risk-averse (someone who avoid risk) based on the clues and available information in long-standing investment plans (Kiran et al., 2020; Raut, 2020). In short, the objective of the optimised stock portfolio is to build a portfolio that offers the lowest level of risk for a desired level of return.

As such, the Quadratic Programming (QP), a nonlinear programming type, has a target function represented by a second-order function to maximise or minimise a quadratic objective function against the linear constraints (Bazaraa et al., 2006). This optimisation tool is frequently used to solve Mean-Variance Optimisation (MVO) by taking into account the mean and variance of stocks in a portfolio investment (Mussafi and Ismail, 2021; Onanye and Agbolamagbin, 2016).

The MVO proposed by Markowitz has been extensively acknowledged as a foundation of modern portfolio theory. Many researchers (Yousfat, 2015; Onanye, A.S. and Agbolamagbin, 2016; Mussafi, 2017) have conducted QP studies of portfolio selection in several stock markets such as Indonesia, Malaysia, Nigeria, and Turkey. Despite its successful utilisation, QP still received criticisms and alternative refined model proposals (Cesarone et al., 2013; Kremer et al., 2018; Škrinjarić and Šego, 2019; Zhang and Zhang, 2011). Zhang and Zhang (2011) propose pivoting-based method while Kremer et al. (2018) introduce eight risk minimization schemes to solve MVO. A study of Limited Asset Markowitz (LAM) also enhances QP by adding a new cardinality constraint that cannot exceed the limited asset (Cesarone et al., 2013). Also, Škrinjarić and Šego (2019) applies Grey Relational Analysis (GRA) which is widely used in ecological, hydrological, geological and similar systems, to build a portfolio compared to Markowitz's efficient frontier. Although many enhancements have been proposed to make the Markowitz model more realistic either exact or heuristic approaches, we analysed one variable that may further improve the optimum result in this study.

This paper aims to construct the Downside Deviation Quadratic Programming (DDQP) as a refinement alternative for QP by introducing downside deviation as a sub-component in the objective function instead of standard deviation. DDQP is expected to minimise risk better than QP. Therefore, we conducted a comparative study to ensure the consistency of the results of the modified model. Apart from minimising the portfolio risk, the constructed model is also expected to result in stock weighting investment.

The contribution of this paper can be summarised as follows: First of all, we extend QP by concerning the asymmetric risk to avoid penalising advantageous (upside) deviations from the geometric mean in the same way as they penalise unfavourable (downside) deviations. The proposed DDQP is then conducted to the active stock index company registered with two popular indices in Indonesia, namely JII (shariah) and IDX30 (conventional). Second, investigating the robustness and consistency of DDQP risk results against QP risk results by constructing multiple scenarios. This step is necessary to ensure that the DDQP carries a lower risk. Finally, verifying the DDQP by employing the heuristic method of PS to approximate the exact optimum risk. The heuristic approach is frequently used to rapidly and efficiently judge the exact result.

The structure of the paper is as follows. Firstly, the literature review explains the stock portfolio, descriptive statistics, downside deviation, optimisation and QP. Which then proceeds with the discussion of the methods used in this study. Next, the results and discussion are presented based on the formulated objectives (DDQP). The results were then compared for consistency with an exact approach of QP and verified using a heuristic approach of Pattern Search (PS). This paper finally ends with a conclusion.

2. Theoretical Framework

2.1. Stock Portfolio

Stock is a security in the ownership of a company. A set of the stock portfolio that complies with Islamic law is referred to as a shariah stock portfolio. However, there are consequences within the label of shariah in stock investment. According to Lusyana and Sherif (2017), investment in stock with fixed income is unacceptable, as it promises a static rate of return and give no right to vote. On the other hand, some financial debt threshold is also used to monitor companies. In particular, the Shariah Advisory Council of Securities Commission Malaysia (SAC-SC) and Dow Jones Islamic Market Indexes (DJIMI) mentioned that the level of debt in companies which included shariah-compliance should not exceed 33% of the market capitalisation from the total equity of a company (Ayedh et al., 2019).

The Financial Services Authority of Indonesia classifies stocks as shariah-compliant if the company declares that they are not involved in (1) Gambling; (2) Conventional financial services; (3) Producing, distributing, trading and/or providing products or services that are prohibited by the National Shariah Board-Majelis Ulama Indonesia (MUI) such as alcohol and pork-related products; (4) Trading of risk that contains gharar or uncertainty (Alam et al., 2017).

It may be argued that these regulation and prohibitions are mainly to protect the interest of all parties involved in the stock market in the objective of shariah or maqasid al-shariah. In addition, it is also important for investors to know that the shariah screening methodology above considers two aspects, i.e., business activities of company and its financial transactions (Ayedh et al., 2019).

In Indonesia, the accepted investment of stocks includes Jakarta Islamic Index (JII), Jakarta Islamic Index 70 (JII70) or Index Saham Syariah Indonesia (ISSI). The investors can use the list of sharia securities (Daftar Efek Syariah) as an investment guide to place managed funds. The list of shariah securities is a collection of securities that are not contrary to shariah principles in the capital market and is officially screened by the National Shariah Board-MUI (Setiawan and Kanila Wati, 2019).

On the other hand, stock portfolios that are not affiliated with shariah principles are known as conventional stock portfolios. There are some conventional stock indices registered in The Indonesia Stock Exchange (IDX) such as LQ45, IDX30, IDX80, IDX BUMN20, etc. IDX provides some indicators to stock indices like market capitalisation, high liquidity and goods fundamental.

2.2. Descriptive Statistics and Some Related Properties

Assuming that a stock has a weekly closed price measurement framework, the returns are measured from the end of the preceding week ($t-1$) to the end of the current week (t). Qoyum et al. (2020) defined geometric return r_t (log-returns) and geometric mean return μ_t for $t = 1, \dots, n$ as follows

$$r_t = \ln \frac{P_t}{P_{t-1}} \quad \text{and} \quad \mu_t = \frac{\sum_{t=1}^n r_t}{n} \quad (1)$$

The standard deviation was used to measure the return change over a period (volatility returns). Suppose r_{it} denotes return for i -th asset (with $i = 1, \dots, N$) at an interval time of $t = 1, \dots, T$ and $\bar{r}_i$ is the mean rate of return. Meanwhile, the standard deviation and the average squared deviation (variance) on the i -th asset (Elton et al., 2014) are:

$$\sigma = \sqrt{\sum_{t=1}^T \frac{(r_{it} - \bar{r}_i)^2}{T}} \quad \text{and} \quad \sigma^2 = \sum_{t=1}^T \frac{(r_{it} - \bar{r}_i)^2}{T} \quad (2)$$

Generally, the variance-covariance matrix has a variance along the diagonal and the covariance appears in the non-diagonal element (Mussafi, 2017). It is defined as follows

$$\text{cov}(R_i, R_j) = \frac{1}{T} \sum_{t=1}^T (r_{it} - \bar{r}_i)(r_{jt} - \bar{r}_j) \quad (3)$$

In terms of portfolio or a set of stocks, let w be weight of stocks in portfolio, μ be expected return of stocks in portfolio, and Σ be variance-covariance matrix. The portfolio expected return and portfolio standard deviation was formulated according to Hartono (2013), respectively, as follows:

$$E(R_p) = w^T \mu \quad \text{and} \quad \sigma_p = (w^T \Sigma w)^{\frac{1}{2}} \quad (4)$$

2.3. Downside Deviation

Unlike standard deviation (a measure of risk) which treats fluctuations above and below the mean in the same pattern, downside deviation accommodates different views of risk. It only observes below the mean or downside part. The calculation of downside deviation depends on the “conditional returns” of the geometric mean; that is, the lowest of either the geometric return minus the geometric mean return or 0. In short, if the return is higher than the geometric mean, the cell indicates a 0, whereas, if the return is lower than the geometric mean, the cell then displays the difference between the two. According to Estrada (2006), the downside deviation of the return on the i -th asset can be defined by avoiding the r_{it} that is greater or equal to $\bar{r}_i$, as depicted below:

$$\hat{\sigma} = \sqrt{\frac{\sum_{t=1}^T [\min(r_{it} - \bar{r}_i, 0)]^2}{T}} \quad (5)$$

2.4. Optimisation and Quadratic Programming

Optimisation is an assortment of methods and algorithms that can be used to find the optimum condition to any problem concerning decision-making. Decision making, whether in the field of engineering or economics, requires the selection of the “best” decision among alternatives. Hence, optimisation theory was developed to choose the best alternative based on a certain objective function from a given feasible region. To solve relatively complex problems, experts suggested the use of nonlinear programming instead of linear programming. The problem is classified as a nonlinear programming problem (NLP) when the objective function is represented as a nonlinear and/or the feasible region is assigned by nonlinear constraints (Chong and Zak, 2013). A class of NLP is formulated as:

$$\begin{aligned} &\text{Minimize } f(x) \\ &\text{subject to } g_i(x) \leq 0 \quad \text{for } i = 1, \dots, m \\ &\quad \quad \quad h_j(x) = 0 \quad \text{for } j = 1, \dots, p \end{aligned} \quad (6)$$

where $x \in \mathbb{R}^n$, $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $g_i: \mathbb{R}^n \rightarrow \mathbb{R}$, $h_j: \mathbb{R}^n \rightarrow \mathbb{R}$ and $m \leq n$. Any point fulfilling the constraints is known as feasible point. The set of all feasible points $\{x \in \mathbb{R}^n : g(x) \leq 0, h(x) = 0\}$ is called the feasible set. Equation 6 must solve feasible points $x_1, \dots, x_n$ to satisfy the constraints

function (both inequality $g_i(x)$ and equality $h_j(x)$) apart from minimising the objective function f .

The following QP exemplifies a particular condition of NLP wherein the objective function is in quadratic and the constraints are linear. According to Bazaraa, Sherali, and Shetty (2006) and Mussafi and Ismail (2020), let c be an n – vector and b be an m – vector. The standard QP is formulated as:

$$\begin{aligned} \min \quad & \frac{1}{2} x^T H x + c^T x \\ \text{s.t.} \quad & A x = b \\ & x \geq 0 \end{aligned} \tag{7}$$

Assuming A is $m \times n$ matrix and H is an $n \times n$ symmetric matrix. The objective function of Equation 7 can be inferred as a convex as the feasible set belongs to polyhedral set and H is a symmetric and positive semi-definite matrix.

2.5. Pattern Search (PS)

PS is rooted in a heuristic method to approach the optimal point performed by the exact method. It is a direct search method used to solve optimisation problems including portfolio optimisation. Firstly, the algorithm begins through an arbitrary initial point to evaluate the objective function. Following the evaluation, PS seeks a set of points nearby the given point called mesh. This current point can be the initial point provided by the user or set from the previous step of the algorithm. The mesh is set by including the current point to a pattern, i.e., a scalar multiple of a set of vectors. If a point in the mesh advances the objective function at the current point, the new point turns into the current point at the following iteration. The evaluation of the objective function and the search for a point towards optimality continues until the algorithm stops when the tolerance constraints are met or a maximum number of iteration is exceeded (Ackora-Prah et al., 2014).

Initialization: Let x_0 be an initial point such that the objective function $f(x_0)$ is finite, and let $M_0 \subset X$ be the mesh identified by a parameter $\Delta_0 > 0$ and x_0

For $k = 0, 1, 2, \dots$ do

1. **SEARCH:** Use some finite strategy seeking an enhanced mesh point, i.e., $x_{k+1} \in M_k$ so that $f(x_{k+1}) < f(x_k)$.
2. **POLL:** If the **SEARCH** step was failed, evaluate f at points in the poll set P_k until an enhanced mesh point x_{k+1} is found.
3. **Update:** If **SEARCH** or **POLL** found an enhanced mesh point, Update x_{k+1} , and set $\Delta_{k+1} \geq \Delta_k$. Otherwise, set $x_{k+1} = x_k$ and set $\Delta_{k+1} < \Delta_k$.

Figure 1: Algorithm for PS

Based on Figure 1, the PS algorithm is separated into two main steps, namely, Search and Poll (Abramson, 2002). A Poll is simply an algorithm that works after a Search. Search attempts to seek an optimal point with an objective function value lower than the current point. If Search has found the optimal point, this point becomes the current point and no Polling is performed

at the iteration. However, failure to determine an optimal point causes the Pattern Search to perform a Poll. The Search method helps to maintain a better local solution and obtain a global solution.

3. Materials and Methods

This chapter discusses the materials and methods used to solve the problems that were proposed in this study. The samples include the historical prices of all stocks listed on the JII (shariah index) and IDX30 (conventional index). Each stock indices have an equal number of members (30 constituents) recorded on IDX from 1st June 2016 to 31st May 2019 (156 weeks).

Constituents in JII must be registered in the review of Daftar Efek Syariah (DES) that is released twice a year, on May and November, by Otoritas Jasa Keuangan (OJK). On the other hand, the members of IDX30 must be registered in the IDX examination review which is disclosed annually in January and July. Based on the aforementioned details, the historical closed prices of the stocks were analysed using descriptive statistics. Furthermore, the portfolio selection was performed by grouping the stocks into portfolio groups through the sectoral approach. The grouping procedure follows the sector classification issued by the authority of IDX.

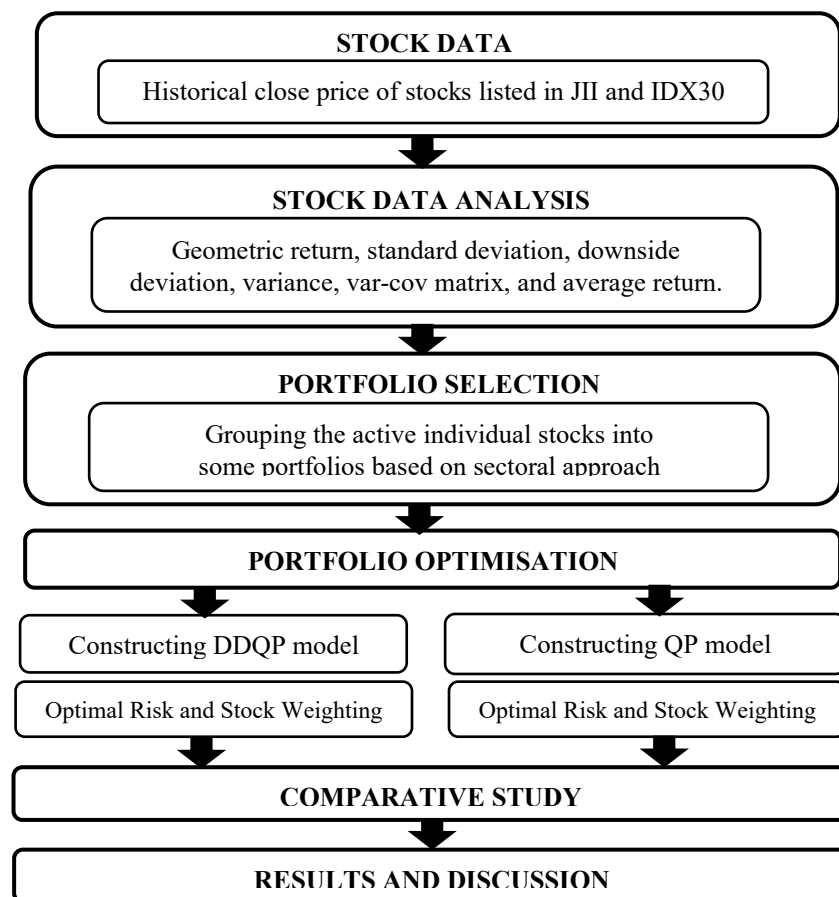


Figure 2: Procedure of Research

Upon the completion of the portfolio selection, portfolio optimisation is performed. Based on Figure 2, this step began with the construction of computer programming for the optimisation

model, i.e., the QP and DDQP models. Moreover, some related variables were substituted into the corresponding models.

The model aims to reduce the level of volatility or risk of stock portfolio investment assuming risk-averse investors. The results of the modified model will be compared with the results of the standard model. Moreover, the algorithm that was constructed in this study also produces stock weighting which corresponds to its objective value. Having said that, short selling or an act that speculates on the decline in a stock price are not allowed. Meanwhile, to evaluate the robustness and consistency of the DDQP model against the QP model, some scenarios were performed. Next, the PS (heuristic method) was conducted to confirm the results of the DDQP model. All computer programming namely QP, DDQP and PS were all implemented using the MATLAB Optimisation Toolbox (MathWorks, 2020).

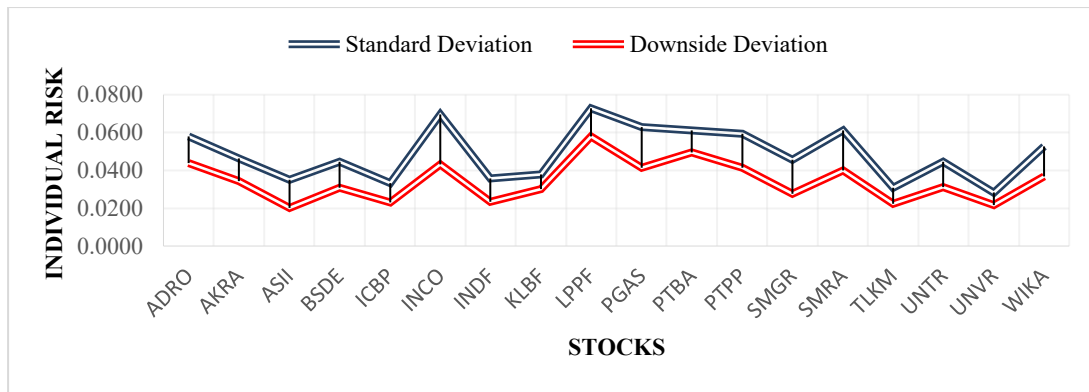
4. Results and Discussion

4.1. Data and Descriptive Statistics

This study used secondary data which was accessed acquiescently through the related website. The dataset contained the weekly data price of active shariah stock index company registered with JII and IDX30 for 156 weeks from 2016 to 2019. The data obtained from Yahoo finance and Investing.com contained actual weekly data and only captured the closing prices. Moreover, the profile, regulation and list of stocks corresponding to JII were retrieved from the OJK and IDX websites. Moreover, some related variables were substituted into the corresponding models.

In total, JII contained 30 stocks. The authority of IDX regularly updates the members of these stocks every six months based on the list of DES released by OJK. In the meantime, the Board of Commissioners of OJK routinely reviews the shariah implementation of all listed stocks. Data collected throughout the 156 weeks would have been through six evaluations based on the IDX release, where only 18 stocks remained consistently active in JII. On the other hand, IDX30 which also comprised of 30 stocks was reviewed by IDX twice a year. During the same period, only 17 active stocks remained active.

Due to the weekly prices of the stocks for 156 weeks, the geometric rate of return (μ_i) was calculated for individual stocks using Equation 1. Meanwhile, the risk measures including the standard deviation (σ_i) and downside deviation ($\hat{\sigma}_i$) were also calculated (Equation 2 and 5). According to Figure 3, the individual risk of downside deviation (red line) of the JII stock index predominantly produced lower risk than that of the standard deviation. Both lines manifested a consistent pattern even though INCO demonstrated the largest gap while UNVR displayed the smallest gap in risk differences compared to the remaining 16 stocks. Having said that, ASII was identified as the stock with the lowest risk in individual risk tests with only 2.013%.



**Figure 3: Comparison of Individual Risk
(Standard Deviation vs. Downside Deviation)**

4.2. Portfolio Selection

During portfolio selection, all stocks were grouped into several portfolios based on a sectoral approach.

Table 1: Portfolio Selection Sector-Based (JII)

No	Code	Business Field	Sectors	Group of Portfolio
1	KLBF	Kalbe Farma Tbk.	Consumer good industry	Group 1
2	ICBP	Indofood CBP Sukses Makmur Tbk.	Consumer good industry	
3	INDF	Indofood Sukses Makmur Tbk.	Consumer good industry	
4	UNVR	Unilever Indonesia Tbk.	Consumer good industry	
5	ADRO	Adaro Energy Tbk.	Mining	Group 2
6	INCO	Vale Indonesia Tbk.	Mining	
7	PTBA	Tambang Batubara Bukit Asam (Persero) Tbk.	Mining	
8	BSDE	Bumi Serpong Damai Tbk.	Property, Real Estate, and Building Construction	Group 3
9	PTPP	PP (Persero) Tbk.	Property, Real Estate, and Building Construction	
10	SMRA	Summarecon Agung Tbk.	Property, Real Estate, and Building Construction	
11	WIKA	Wijaya Karya (Persero) Tbk.	Property, Real Estate, and Building Construction	
12	AKRA	AKR Corporindo Tbk.	Trade, service, and investment	Group 4
13	UNTR	United Tractors Tbk.	Trade, service, and investment	
14	LPPF	Matahari Department Store Tbk.	Trade, service, and investment	
15	PGAS	Perusahaan Gas Negara (Persero) Tbk.	Infrastructure, utilities, and transportation	Group 5
16	TLKM	Telekomunikasi Indonesia (Persero) Tbk.	Infrastructure, utilities, and transportation	
17	SMGR	Semen Indonesia (Persero) Tbk.	Basic industry and chemical	
18	ASII	Astra International Tbk.	Miscellaneous industry	

Table 2: Portfolio Selection Sector-Based (IDX30)

No	Code	Business Field	Sectors	Group of Portfolio
1.	GGRM	Gudang Garam Tbk.	Consumer good industry	Group 6
2.	ICBP	Indofood CBP Sukses Makmur Tbk.	Consumer good industry	
3.	INDF	Indofood Sukses Makmur Tbk.	Consumer good industry	
4.	KLBF	Kalbe Farma Tbk.	Consumer good industry	

5.	UNVR	Unilever Indonesia Tbk.	Consumer good industry	
6.	BBCA	Bank Central Asia Tbk.	Finance	
7.	BBNI	Bank Negara Indonesia (Persero) Tbk.	Finance	
8.	BBRI	Bank Rakyat Indonesia (Persero) Tbk.	Finance	Group 7
9.	BBTN	Bank Tabungan Negara (Persero) Tbk.	Finance	
10.	BMRI	Bank Mandiri (Persero) Tbk.	Finance	
11.	PGAS	Perusahaan Gas Negara (Persero) Tbk.	Infrastructure, utilities, and transportation	
12.	TLKM	Telekomunikasi Indonesia (Persero) Tbk.	Infrastructure, utilities, and transportation	Group 8
13.	INTP	Indocement Tunggal Prakarsa Tbk.	Basic industry and chemical	
14.	SMGR	Semen Indonesia (Persero) Tbk.	Basic industry and chemical	
15.	ADRO	Adaro Energy Tbk.	Mining	
16.	ASII	Astra International Tbk.	Miscellaneous industry	Group 9
17.	UNTR	United Tractors Tbk.	Trade, service, and investment	

Formally, the IDX authorities classified all the stocks under shariah and conventional into 9 sectors: (1) Agriculture, (2) Basic Industry and Chemicals, (3) Consumer Good Industry, (4) Finance, (5) Infrastructure, Utilities and Transportation, (6) Mining, (7) Miscellaneous Industry, (8) Property, Real Estate and Building Construction, and (9) Trade Service Investment.

Based on Table 1, the JII grouping for some standalone stocks like SMGR and ASII in their respective sectors was to be merged into group 5. Moreover, some standalone stocks on the stock index of IDX30 were also merged (Table 2).

4.3. Portfolio Optimisation

The main contribution of this study is in the construction of the optimisation model for a stock portfolio called the DDQP. This model is an improvement from the existing QP model, particularly with a change in the variable of the objective function. The conventional standard deviation was substituted by the downside deviation in this new model. This substitution can cause variance whereby the objective function becomes the square of downside deviation and is no longer the square of standard deviation. Hence, the optimisation problem of Equation 7 was accustomed to portfolio optimisation based on the objective and constraint functions.

Assuming stocks $S_1, S_2, \dots, S_n$ for $n \geq 2$ with random variable of returns. Let $\hat{\mu}_i$ be the geometric mean return of stocks S_i . In the calculation, $\hat{\mu}_i$ allows only the r_{it} that are smaller or equal to $\bar{r}_i$. Let $\hat{\sigma}_i$ be downside deviation of stocks S_i and let x_i be the proportion of the amount invested in the portfolio (set of stocks). As $i \neq j$, ρ_{ij} indicates the correlation coefficient of the returns of stocks S_i and S_j . Also, let $\hat{\mu} = [\hat{\mu}_1, \dots, \hat{\mu}_n]^T$ and $H = (\hat{\sigma}_{ij})$ be the $n \times n$ symmetric variance-covariance matrix with $\hat{\sigma}_{ii} = \hat{\sigma}_i^2$ and $\hat{\sigma}_{ij} = \rho_{ij} \hat{\sigma}_i \hat{\sigma}_j$, respectively, for $i \neq j$. Meanwhile, the expected return and variance of the outcome portfolio $x = (x_1, \dots, x_n)$ were estimated as follows:

$$\begin{aligned}
E[x] &= x_1 \hat{\mu}_1 + x_2 \hat{\mu}_2 + \dots + x_n \hat{\mu}_n = \hat{\mu}^T x \text{ and} \\
Var[x] &= \sum_{i,j} \rho_{ij} \hat{\sigma}_i \hat{\sigma}_j x_i x_j = x^T H x, \\
\text{where } \rho_{ii} &\equiv 1.
\end{aligned} \tag{8}$$

It is important to note that the variance is nonnegative at all times since it is the expectation of a nonnegative random variable. It implies that $x^T H x \geq 0$ for any x , where H is positive semi-definite. However, it will be assumed as positive definite, that is fundamentally correspondent to no repetitive assets in group of $S_1, S_2, \dots, S_n$. Also the set of acceptable portfolio is assumed to be a nonempty polyhedral set and is denoted as $X := \{x : Ax = b, Cx \geq d\}$, where A is $m \times n$ matrix, b is m -dimensional vector, C is $p \times n$ matrix and d is p -dimensional vector.

Whereas, constraint in the set X can be expressed as $\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n = 1$.

Since it is assumed that the variance H positive definite is strictly a convex function of the portfolio variables, it leads to a unique portfolio X with minimum variance. Let's denote this portfolio with $x_{\min}$ and its return $\hat{\mu}^T x_{\min}$ with R , to make $x_{\min}$ an efficient portfolio. The goal is to obtain a minimum risk portfolio of the given stocks to achieve at least an equivalent estimated expected return. Mathematically, this formulation in consort with Equation 8 yields a DDQP for portfolio optimisation:

$$\begin{aligned}
& \min_x \frac{1}{2} x^T H x + c^T x \\
& s.t. \quad \hat{\mu}^T x \geq R \\
& \quad \sum_{i=1}^n x_i = 1 \\
& \quad x \geq 0
\end{aligned} \tag{9}$$

According to equation 9, the objective function refers to variance-covariance matrix. It corresponds to (one half) the total variance of the portfolio. The constant $\frac{1}{2}$ is added for convenience in the optimality conditions so that it does not affect the optimal solution. Having said that, the first constraint was expressed by geometric mean return with an expected return greater than or equal to the estimated value of R , while the second constraint inferred that the sum of all solutions must be 1. Therefore, in short, DDQP is essentially a modification of several variable inputs.

However, we employed the interior point method to manage non-linear programming (NLP). Figure 4 illustrates the basic pseudo-code of the interior point method (Schmidt, 2015). Schmidt argued that the interior point method should be replaced by a series of barrier functions with implications in terms of inequality constraints for the use of the primal-dual system. The interior-point method ends when the Karush–Kuhn–Tucker (KKT) point was approximately reached.

Input: Starting point $(x^{(0)}, \lambda^{(0)}, z^{(0)})$ with $x^{(0)}, z^{(0)} > 0$, initial barrier parameter $\mu^{(0)} > 0$, convergence tolerance $\varepsilon > 0$.
Set $k = 0$.
while $e^{(k)} > \varepsilon$ **do**

Compute the search direction $(\Delta x^{(k)}, \Delta \lambda^{(k)}, \Delta z^{(k)})$ by solving KKT system.
 Compute maximum step lengths $\bar{\alpha}_{pri}^{(k)}, \bar{\alpha}_{dual}^{(k)}$ using fraction-to-the-boundary rule.
 Apply a globalization strategy to acquire actual step lengths $\alpha_{pri}^{(k)}, \alpha_{dual}^{(k)}$.

$$x^{(k+1)} = x^{(k)} + \alpha_{pri}^{(k)} \Delta x^{(k)},$$

 Compute the new iterate $\lambda^{(k+1)} = \lambda^{(k)} + \alpha_{dual}^{(k)} \Delta \lambda^{(k)},$

$$z^{(k+1)} = z^{(k)} + \alpha_{dual}^{(k)} \Delta z^{(k)}.$$

 Compute a new barrier parameter $\mu^{(k+1)}$ or set $\mu^{(k+1)} = \mu^k$.
 Set $k \leftarrow k + 1$.

Figure 4: Algorithm of Interior Point for NLP

On the other hand, the construction of standard QP is almost similar to Equation 9. Let μ_i be a geometric mean return of stocks S_i and σ_i be the standard deviation of stocks S_i . For $i \neq j$, ρ_{ij} indicates the correlation coefficient of the returns of stocks S_i and S_j . Let $\mu = [\mu_1, \dots, \mu_n]^T$ and $Q = (\sigma_{ij})$ be the $n \times n$ symmetric variance-covariance matrix with $\sigma_{ii} = \sigma_i^2$ and $\sigma_{ij} = \rho_{ij} \sigma_i \sigma_j$ for $i \neq j$. The expected return and the variance of the outcome portfolio x were $E[x] = x_1 \mu_1 + x_2 \mu_2 + \dots + x_n \mu_n = \mu^T x$ and $Var[x] = \sum_{i,j} \rho_{ij} \sigma_i \sigma_j x_i x_j = x^T Q x$, respectively. The assumptions for QP (i.e., the set of admissible portfolio, positive definite, and efficient portfolio) are analogous to DDQP, hence, the QP model for portfolio optimisation is:

$$\begin{aligned}
 & \min_x \quad \frac{1}{2} x^T Q x + c^T x \\
 & s.t. \quad \mu^T x \geq R, \\
 & \quad \sum_{i=1}^n x_i = 1, \quad x \geq 0
 \end{aligned} \tag{10}$$

4.4. Numerical Analysis

The two models of portfolio optimisation (DDQP and QP) were tested on portfolio group 1 as a sample. According to Table 1, group 1 consists of five stocks, namely KLBFI, ICBP, INDF and UNVR. Firstly, the DDQP model was used to analyse the chosen test case using the weekly closed prices of all stocks in group 1 and its descriptive statistics. Moreover, the geometric mean return $\hat{\mu}$ was obtained from the data (Equation 1) with some limitation as mentioned in the assumption of obtaining Equation 8. Such limitation will yield the negative value since $\hat{\mu}$ represent a negative return. Suppose its upper bound is expected return ($R = 0.05$), the variance-covariance matrix $H_{4 \times 4}$ can be calculated using Equation 3.

$$\hat{\mu} = [-0.0289015 \quad -0.02014335 \quad -0.02797985 \quad -0.021982307] \quad \text{and}$$

$$H_{4 \times 4} = \begin{bmatrix} 0.000872068 & 0.000288324 & 0.000433159 & 0.000429531 \\ 0.000288324 & 0.00047927 & 0.000382812 & 0.000208687 \\ 0.000433159 & 0.000382812 & 0.000670215 & 0.00031416 \\ 0.000429531 & 0.000208687 & 0.00031416 & 0.000474304 \end{bmatrix}$$

Assuming that KLBF, ICBP, INDF, and UNVR are denoted by variables x_A, x_B, x_C , and x_D . By substituting $H_{4 \times 4}$, $\hat{\mu}$ and R into Equation 9, the portfolio optimisation of DDQP can be constructed as follows:

$$\begin{aligned} \min_x \quad & 0.00087206x_A^2 + 0.0004792x_B^2 + 0.0006702x_C^2 + 0.0004743x_D^2 + \\ & (2 \times 0.00028832x_Ax_B) + (2 \times 0.00043315x_Ax_C) + (2 \times 0.00042953x_Ax_D) + \\ & (2 \times 0.00038281x_Bx_C) + (2 \times 0.00020868x_Bx_D) + (2 \times 0.00031416x_Cx_D) \\ \text{s.t.} \quad & -0.0289015x_A - 0.02014335x_B - 0.02797985x_C - 0.021982307x_D \geq 0.05 \\ & x_A + x_B + x_C + x_D = 1 \\ & x_A, x_B, x_C, x_D \geq 0 \end{aligned} \quad (11)$$

The mathematical model for Equation 11 can be seen from the perspective of financial engineering. Hence, it can be technically solved using a numerical computing environment like the MATLAB software. The construction of computer programming utilises the m-file MATLAB through an optimisation toolbox, particularly, optimoptions and quadprog. Accordingly, the scripts of m-file admit input arguments and yield output arguments (Brandimarte, 2013; MathWorks, 2020). The data input contains of the number of stocks, variance-covariance matrix, transpose of geometric mean, and expected return. The declared input was supported by optimisation operator which then produced the program output, i.e., the minimum value of risk and stock weighting. The numerical results of DDQP of portfolio group 1 were obtained by running the proposed m-file program.

-- DDQP for Portfolio of Sector One --				Minimum found that satisfies the constraints.
Please fill in the number of stocks: 4				<stopping criteria details>
-----				-----
INPUT				OUTPUT
-----				-----
Variance-covariance matrix:				The best function value found was: 0.000171368
1.0e-03 *				Minimum value of risk (SD): 1.309077%
0.8721	0.2883	0.4332	0.4295	Stock weighting (feasible set):
0.2883	0.4793	0.3828	0.2087	0.0058%
0.4332	0.3828	0.6702	0.3142	49.4860%
0.4295	0.2087	0.3142	0.4743	0.0796%
				50.4286%
Geometric mean:				Algorithm: interior-point-convex
-0.0289	-0.0201	-0.0280	-0.0220	The number of iterations: 7
Expected return:				The number of function evaluations was : 70
0.0500				The algorithm converged to the local min x.
-----				-----
				Elapsed time is 1.298480 seconds.

Figure 5: Output of DDQP m-file Program (MATLAB)

Figure 5 demonstrates the minimum value of risk and the asset allocation of investment for each stock on group 1 portfolio (sector of consumer good industry). Assume investors or portfolio managers have 100 million IDR to invest in such a portfolio. Using the DDQP exact method, they will carry the risk of 1.3091% (1.309 million IDR) on this portfolio. It also can be inferred that UNVR and ICBP hold the two biggest fund allocation to be considered with 50.4286% (50.428 million IDR) and 49.4860% (49.486 million IDR), respectively. Followed by INDF with 0.08% (0.08 million IDR) and KLBF with 0.006% (0.006 million IDR) of the total fund.

Equation 10 represents the numerical analysis of the QP model for all portfolio groups. To evaluate the performance of the two models (DDQP and QP), the portfolio risks of all portfolio groups estimated using the two finance optimisation models were examined.

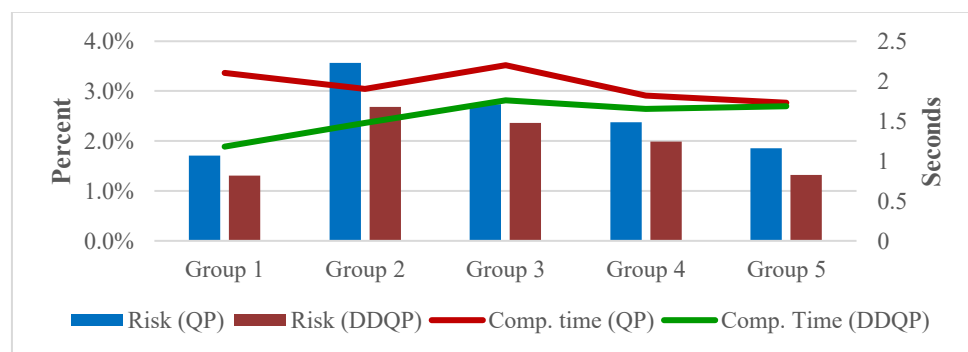


Figure 6: Comparison of the Minimum Risk among all Portfolio Groups generated by DDQP and QP (JII)

Comparably, the minimum value of risk generated by QP was higher than that of DDQP (Figure 6). Based on the figure, group 1 indicated the lowest risk among other portfolio groups with a risk level of 1.3091% for DDQP and 1.7121% for QP. The red and green lines within the figure indicate the computation time taken to execute each portfolio group using QP and DDQP. Overall, the DDQP model required less computation time compared to the QP model.

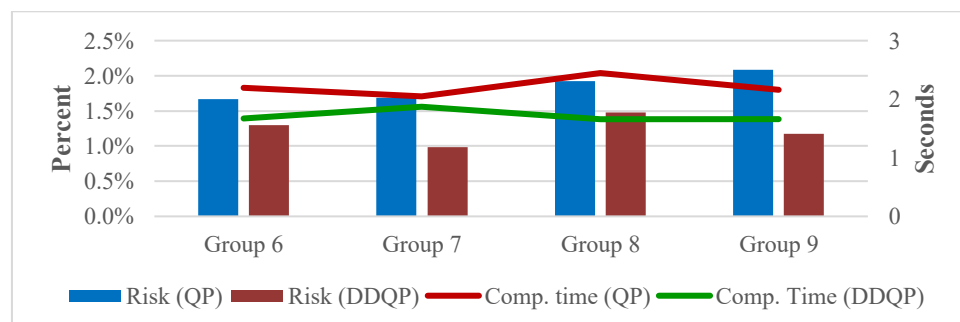


Figure 7: Comparison of the Minimum Risk among all Portfolio Groups generated by DDQP and QP (IDX30)

Analogous to the procedure above, portfolio optimisation of IDX30 corresponded to the group arrangement in Table 2 examined using DDQP and QP. Figure 7 revealed that group 7 representing the finance sector yielded the lowest risk using DDQP with a risk value of 0.9859%. Hence, it was considered “the best” portfolio in the IDX30. Meanwhile, the green line indicated that all execution times of DDQP were shorter than that of the QP.

In short, the examination of both the models only provided the portfolio risk but also produced results on stock weighting. Stock weighting is an essential instrument for potential investors in determining the percentage of stock investment in a portfolio. The stacked area and stacked line chart in Figure 8 provide information on the weight of stocks for each portfolio groups generated by DDQP and QP. However, the total weighting of each portfolio groups should not exceed 100%. For example, the weighting of four stocks according to portfolio group 1 for the DDQP model was 0.006%, 49.48%, 0.08%, and 50.42% for KLBFI, ICBP, INDF and UNVR, respectively. Meanwhile, the weighting of four stocks in the same group using the QP model was more balanced in terms of proportion, namely KLBFI by 10.68%, ICBP by 26.45%, INDF by 13.68% and UNVR by 50.42%.

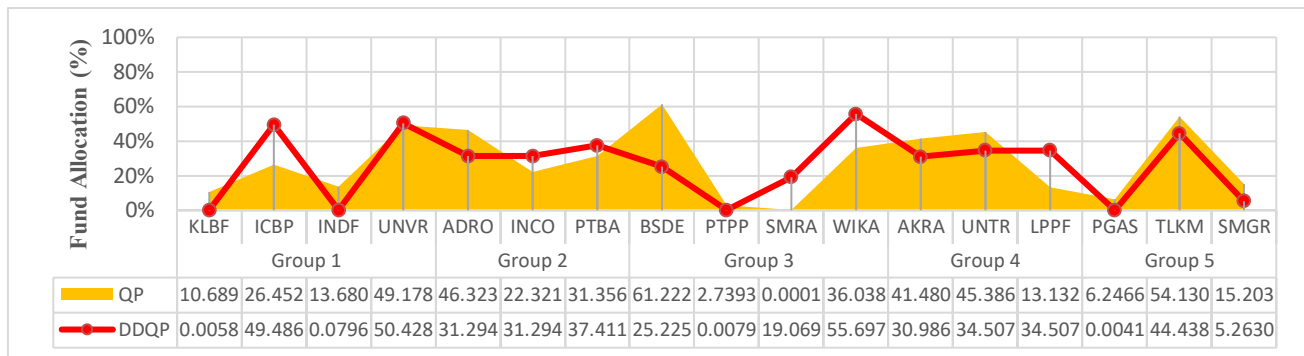


Figure 8: Comparison of the Weights of Stocks among all Portfolio Groups generated by DDQP and QP (JII)

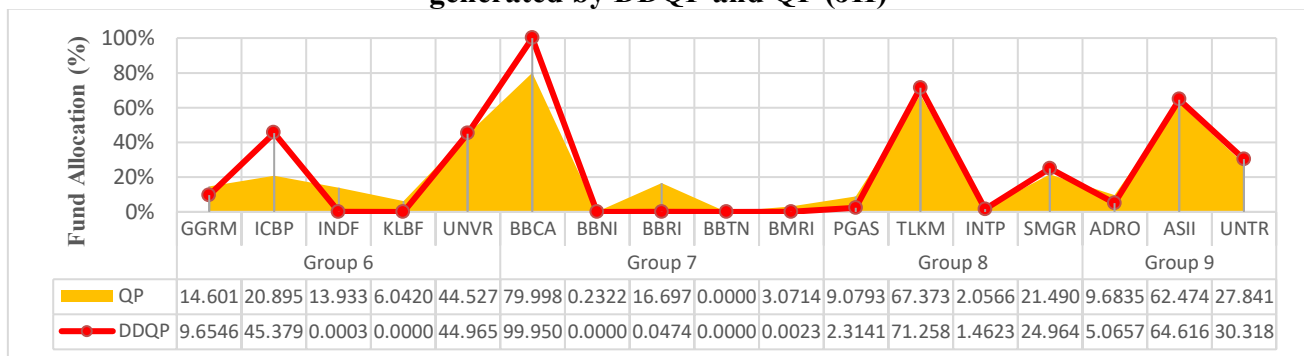


Figure 9: Comparison of the Weights of Stocks among all Portfolio Groups generated by DDQP and QP (IDX30)

Similarly, Figure 9 depict the stock weighting for IDX30 generated by DDQP and QP. The stock weighting of DDQP of portfolio group 9 was almost similar to the stock weighting of QP as well as the weighting of portfolio group 8.

4.5. Robustness and Consistency

On the other hand, to examine the robustness and consistency of DDQP against QP, a comparative study for both syariah and conventional stock indices was performed by drastically changing one or more of the random input stocks in certain portfolios (Lu and White,

2014). The implementation of this empirical analysis was translated into the following scenarios:

- (1) scenario A represents portfolio group 1 (the best portfolio of shariah JII);
- (2) scenario B denotes the changing of portfolio group 1 by removing an arbitrary stock, i.e., UNVR;
- (3) scenario C denotes the changing of portfolio group 1 by removing two arbitrary stocks, i.e., UNVR and INDF;
- (4) scenario D characterises the changing of portfolio group 1 by adding an arbitrary stock from another portfolio, i.e., ADRO;
- (5) scenario E characterises the changing of portfolio group 1 by adding two arbitrary stocks from another portfolio, i.e., ADRO and BSDE;
- (6) scenario F represents portfolio group 7 (the best portfolio of conventional IDX30);
- (7) scenario G denotes the changing of portfolio group 7 by removing an arbitrary stock, i.e., BBKA;
- (8) scenario H denotes the changing portfolio group 7 by removing two arbitrary stock, i.e., BBKA and BBNI;
- (9) scenario I characterises the changing of portfolio group 7 by adding an arbitrary stock from another portfolio, i.e., INTP;
- (10) scenario J characterises the changing portfolio group 7 by adding two arbitrary stocks from another portfolio, i.e., INTP and GGRM.

Table 3 summarises the results of portfolio risk (in percentage) and its computation time (in seconds) for all scenarios generated by the DDQP and QP models. Comparatively, scenario E yielded the lowest risk among the shariah portfolio, while scenario J produced the lowest risk among the conventional portfolio. Meanwhile, scenario A demonstrated the fastest execution time. Hence, it can be inferred that the results of risk and computation time generated by DDQP are always consistent in producing better results than QP.

Table 3: Risk Level and Computation Time of DDQP and QP Model based on Scenarios

Stock Indices	Scenarios	Set of Stocks (Portfolio)	DDQP		QP	
			Risk level	Comp. time	Risk level	Comp. time
Shariah (JII)	Scenario A	KLBF, ICBP, INDF, UNVR	1.3091%	1.2984	1.7121%	2.1017
	Scenario B	KLBF, ICBP, INDF	1.4625%	1.2208	1.9501%	2.0608
	Scenario C	KLBF and ICBP	1.470%	1.6231	2.0022%	2.34801
	Scenario D	KLBF, ICBP, INDF, UNVR, ADRO	1.2947%	2.1914	1.6520%	2.5237
	Scenario E	KLBF, ICBP, INDF, UNVR, ADRO, BSDE	1.2922%	1.5749	1.6509%	2.4829
Conventional (IDX30)	Scenario F	BBKA, BBNI, BBRI, BBTN, BMRI	0.9859%	1.8715	1.6858%	2.0487
	Scenario G	BBNI, BBRI, BBTN, BMRI	1.6074%	1.7511	2.2671%	2.1449
	Scenario H	BBRI, BBTN, BMRI	1.6073%	1.6065	2.2798%	2.3712
	Scenario I	BBKA, BBNI, BBRI, BBTN, BMRI, INTP	0.9859%	1.5893	1.4829%	1.9968
	Scenario J	BBKA, BBNI, BBRI, BBTN, BMRI, INTP, GGRM	0.9829%	1.8672	1.6342%	2.3482

Finally, to confirm the performance of the proposed method (DDQP), 10 scenarios were experimented using a heuristic method (PS) to approximate the optimum solution (Khan et al., 2020). Then, the results obtained between DDQP and PS were compared (Table 4). The comparison included risk (square root of optimal value), optimal value (best function value), number of iterations, number of function evaluations and computation time. Based on the observation, the arbitrary initial point x_0 , the optimal value and the risk level of PS for almost all the cases were relatively approximate to the optimal value and the risk level generated by DDQP.

The computational complexity of PS was discussed by Aamir et al. (2011) as the logarithmic time or $O(\log n)$, while the computational complexity of DDQP correlated with the interior point method of a cubic time or $O(n^3)$ (Ye and Tse, 1989). Moreover, Khan et al. (2020) declared that the computational complexity of the optimisation algorithm can be represented as the number of evaluations of the objective function. Therefore, DDQP outperformed PS in five scenarios (A, B, C, E and H) based on the number of objective function evaluations (Table 4).

Table 4: Comparison of DDQP and PS for various Scenarios

	Scenario A		Scenario B		Scenario C		Scenario D		Scenario E	
	DDQP	PS	DDQP	PS	DDQP	PS	DDQP	PS	DDQP	PS
Risk	1.309077	1.309078	1.462489	1.461027	1.470043	1.468574	1.294654	1.295392	1.292153	1.294685
Optimal Value	0.0001714	0.0001714	0.0002139	0.0002135	0.0002161	0.0002157	0.0001676	0.0001678	0.0001670	0.0001676
Iterations	7	24	6	72	6	50	7	32	7	82
Func. Evaluat.*	70	79	60	362	24	190	120	119	266	329
Comp. Time**	1.2985	4.6599	1.2208	6.4647	1.6231	4.4225	2.1914	3.5205	1.5749	7.2332

	Scenario F		Scenario G		Scenario H		Scenario I		Scenario J	
	DDQP	PS	DDQP	PS	DDQP	PS	DDQP	PS	DDQP	PS
Risk	0.985888	0.984915	1.607367	1.634341	1.607341	1.634341	0.985886	0.984915	0.982907	0.984915
Optimal Value	0.0000972	0.0000970	0.0002584	0.0002671	0.0002584	0.0002671	0.0000972	0.0000970	0.0000966	0.0000970
Iterations	8	24	6	24	6	60	8	60	7	24
Func. Evaluat.*	144	84	110	73	60	123	203	186	272	108
Comp. Time**	1.8715	3.2072	1.7511	4.7274	1.6065	6.0795	1.5893	7.2003	1.8672	3.2036

*Risk is standard deviation of optimal value (%); *Total number of function evaluations; **Computation time is in seconds as measured in MATLAB.*

As for the number of iterations, all cases indicated that PS required more iterations than DDQP. Moreover, DDQP was faster in algorithm computation time compared to PS. All the tests were performed in MATLAB which provided a genuine implementation of DDQP and PS, hence, the comparison of computation time was relatively fair.

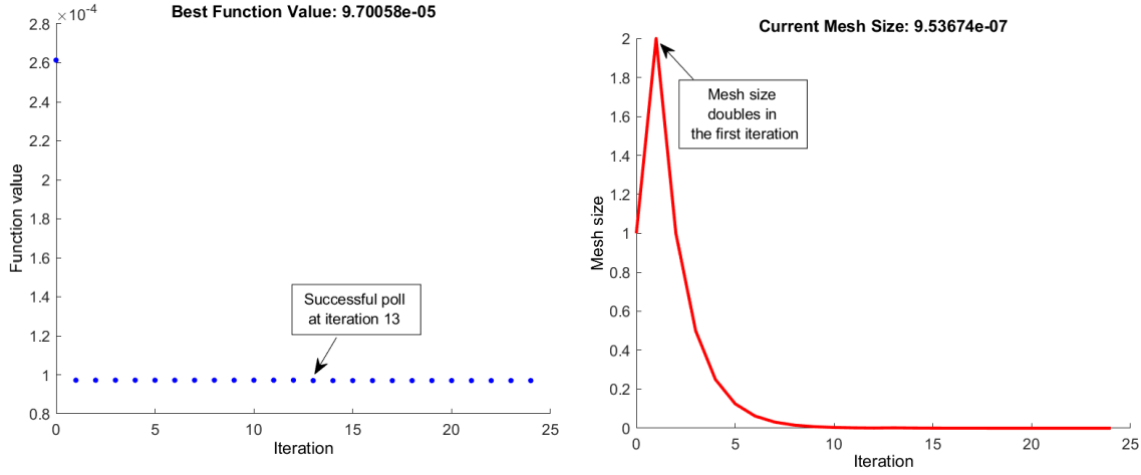


Figure 10: (a) Convergence of PS Optimal Value in Scenario J (left) and (b) Convergence of PS Mesh Size in Scenario J (right)

Moreover, referring to scenario J with the least number of iterations among all scenarios, Figure 10a displayed the convergence of optimal value by PS. The scenario only needed 13 iterations to obtain the optimal value. Conversely, PS was permitted to maintain the search in the neighbourhood of the optimal point to raise the conviction level of the result. It took 11 more iterations for PS to stop and return to the optimal value.

Figure 10b displays that it only took one iteration for the Poll to be successful as the mesh size kept increasing since the algorithm expanded the scope of the search. This scenario continued until iteration number 13 where the mesh size reached 0.001953. Having said that, the refinement of the mesh continued until the mesh size was less than the tolerance of the mesh.

5. Conclusion

The downside deviation which focuses solely on the downside volatility provided a more volatile individual risk than the standard deviation. Such volatility could occur due to the rate below the average return, whereby all returns were either negative or zero. This fact was also confirmed by the individual risk test values of all active stocks that were listed in JII and IDX30 from June 2016 to May 2019. Hence, it can be concluded that all the stocks that were evaluated using downside deviation offered lower individual risk than those tested with the standard deviation, even most of them have a significant difference.

We have constructed an alternative model so-called Downside Deviation Quadratic Programming (DDQP) to solve classical MVO problem initiated by Markowitz. This model offers downside deviation instead of standard deviation. Referring to the test cases above both shariah and conventional stock indices (portfolio group 1-9), our proposed model is able to reduce the level of portfolio risk better than that generated using Quadratic Programming (QP) with average of efficiency ratio of 78.52% for JII and 67.37% for IDX30. In addition, the DDQP model can also identify the weight of stocks in the portfolio as the QP model. Likewise, the output of ten scenarios that consider both shariah and conventional stock indices also exhibited that DDQP model is robust and consistently stable in providing more efficient risk of portfolio and running time. These findings may be used as an alternative way for investors and portfolio managers to construct promising portfolios with low risk according to their

respective preferences, whether shariah or conventional stocks. Finally, the heuristic method of Pattern Search (PS) Algorithm applied to ten scenarios succeeded in justifying the results of DDQP with a tiny average gap of 1.74953E-06.

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